SOLUTION OF HOMEWORK 04

1.

IDENTIFY: $s = r\theta$, with θ in radians.

SET UP: π rad $= 180°$.

EXECUTE: **(a)** $\theta = \dfrac{s}{r} = \dfrac{1.50 \text{ m}}{2.50 \text{ m}} = 0.600 \text{ rad} = 34.4°$

(b) $r = \dfrac{s}{\theta} = \dfrac{14.0 \text{ cm}}{(128°)(\pi \text{ rad}/180°)} = 6.27 \text{ cm}$

(c) $s = r\theta = (1.50 \text{ m})(0.700 \text{ rad}) = 1.05 \text{ m}$

EVALUATE: An angle is the ratio of two lengths and is dimensionless. But, when $s = r\theta$ is used, θ must be in radians. Or, if $\theta = s/r$ is used to calculate θ, the calculation gives θ in radians.

2.

IDENTIFY: $\theta - \theta_0 = \omega t$, since the angular velocity is constant.

SET UP: 1 rpm $= (2\pi/60)$ rad/s.

EXECUTE: **(a)** $\omega = (1900)(2\pi \text{ rad}/60 \text{ s}) = 199 \text{ rad/s}$

(b) $35° = (35°)(\pi/180°) = 0.611$ rad. $t = \dfrac{\theta - \theta_0}{\omega} = \dfrac{0.611 \text{ rad}}{199 \text{ rad/s}} = 3.1 \times 10^{-3} \text{ s}$

EVALUATE: In $t = \dfrac{\theta - \theta_0}{\omega}$ we must use the same angular measure (radians, degrees or revolutions) for both $\theta - \theta_0$ and ω.

3.

IDENTIFY: Apply the constant angular acceleration equations.

SET UP: Let the direction the wheel is rotating be positive.

EXECUTE: **(a)** $\omega_z = \omega_{0z} + \alpha_z t = 1.50 \text{ rad/s} + (0.300 \text{ rad/s}^2)(2.50 \text{ s}) = 2.25 \text{ rad/s}$.

(b) $\theta - \theta_0 = \omega_{0z} t + \frac{1}{2}\alpha_z t^2 = (1.50 \text{ rad/s})(2.50 \text{ s}) + \frac{1}{2}(0.300 \text{ rad/s}^2)(2.50 \text{ s})^2 = 4.69 \text{ rad}$.

EVALUATE: $\theta - \theta_0 = \left(\dfrac{\omega_{0z} + \omega_z}{2}\right)t = \left(\dfrac{1.50 \text{ rad/s} + 2.25 \text{ rad/s}}{2}\right)(2.50 \text{ s}) = 4.69 \text{ rad}$, the same as calculated with another equation in part (b).

4.

IDENTIFY: Apply the constant angular acceleration equations to the motion of the fan.

(a) SET UP: $\omega_{0z} = (500 \text{ rev/min})(1 \text{ min}/60 \text{ s}) = 8.333 \text{ rev/s}$, $\omega_z = (200 \text{ rev/min})(1 \text{ min}/60 \text{ s}) = 3.333 \text{ rev/s}$, $t = 4.00 \text{ s}$, $\alpha_z = ?$

$\omega_z = \omega_{0z} + \alpha_z t$

EXECUTE: $\alpha_z = \dfrac{\omega_z - \omega_{0z}}{t} = \dfrac{3.333 \text{ rev/s} - 8.333 \text{ rev/s}}{4.00 \text{ s}} = -1.25 \text{ rev/s}^2$

$\theta - \theta_0 = ?$

$\theta - \theta_0 = \omega_{0z}t + \frac{1}{2}\alpha_z t^2 = (8.333 \text{ rev/s})(4.00 \text{ s}) + \frac{1}{2}(-1.25 \text{ rev/s}^2)(4.00 \text{ s})^2 = 23.3 \text{ rev}$

(b) SET UP: $\omega_z = 0$ (comes to rest); $\omega_{0z} = 3.333 \text{ rev/s}$; $\alpha_z = -1.25 \text{ rev/s}^2$; $t = ?$

$\omega_z = \omega_{0z} + \alpha_z t$

EXECUTE: $t = \dfrac{\omega_z - \omega_{0z}}{\alpha_z} = \dfrac{0 - 3.333 \text{ rev/s}}{-1.25 \text{ rev/s}^2} = 2.67 \text{ s}$

EVALUATE: The angular acceleration is negative because the angular velocity is decreasing. The average angular velocity during the 4.00 s time interval is 350 rev/min and $\theta - \theta_0 = \omega_{\text{av-}z}t$ gives $\theta - \theta_0 = 23.3 \text{ rev}$, which checks.

5.

IDENTIFY: Use a constant angular acceleration equation and solve for ω_{0z}.

SET UP: Let the direction of rotation of the flywheel be positive.

EXECUTE: $\theta - \theta_0 = \omega_{0z}t + \frac{1}{2}\alpha_z t^2$ gives $\omega_{0z} = \dfrac{\theta - \theta_0}{t} - \frac{1}{2}a_z\, t = \dfrac{60.0 \text{ rad}}{4.00 \text{ s}} - \frac{1}{2}(2.25 \text{ rad/s}^2)(4.00 \text{ s}) = 10.5 \text{ rad/s}.$

EVALUATE: At the end of the 4.00 s interval, $\omega_z = \omega_{0z} + \alpha_z t = 19.5 \text{ rad/s}$.

$$\theta - \theta_0 = \left(\frac{\omega_{0z} + \omega_z}{2}\right)t = \left(\frac{10.5 \text{ rad/s} + 19.5 \text{ rad/s}}{2}\right)(4.00 \text{ s}) = 60.0 \text{ rad}, \text{ which checks.}$$

6.

IDENTIFY: The linear distance the elevator travels, its speed and the magnitude of its acceleration are equal to the tangential displacement, speed and acceleration of a point on the rim of the disk. $s = r\theta$, $v = r\omega$ and $a = r\alpha$. In these equations the angular quantities must be in radians.

SET UP: $1 \text{ rev} = 2\pi \text{ rad}$. $1 \text{ rpm} = 0.1047 \text{ rad/s}$. $\pi \text{ rad} = 180°$. For the disk, $r = 1.25 \text{ m}$.

EXECUTE: **(a)** $v = 0.250 \text{ m/s}$ so $\omega = \dfrac{v}{r} = \dfrac{0.250 \text{ m/s}}{1.25 \text{ m}} = 0.200 \text{ rad/s} = 1.91 \text{ rpm}.$

(b) $a = \frac{1}{8}g = 1.225 \text{ m/s}^2$. $\alpha = \dfrac{a}{r} = \dfrac{1.225 \text{ m/s}^2}{1.25 \text{ m}} = 0.980 \text{ rad/s}^2.$

(c) $s = 3.25 \text{ m}$. $\theta = \dfrac{s}{r} = \dfrac{3.25 \text{ m}}{1.25 \text{ m}} = 2.60 \text{ rad} = 149°.$

EVALUATE: When we use $s = r\theta$, $v = r\omega$ and $a_{\tan} = r\alpha$ to solve for θ, ω and α, the results are in rad, rad/s and rad/s^2.

7.

IDENTIFY: Use constant acceleration equations to calculate the angular velocity at the end of two revolutions. $v = r\omega$.

SET UP: 2 rev $= 4\pi$ rad. $r = 0.200$ m.

EXECUTE: **(a)** $\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$. $\omega_z = \sqrt{2\alpha_z(\theta - \theta_0)} = \sqrt{2(3.00\text{ rad/s}^2)(4\pi\text{ rad})} = 8.68$ rad/s.

$a_{\text{rad}} = r\omega^2 = (0.200\text{ m})(8.68\text{ rad/s})^2 = 15.1\text{ m/s}^2$.

(b) $v = r\omega = (0.200\text{ m})(8.68\text{ rad/s}) = 1.74$ m/s. $a_{\text{rad}} = \dfrac{v^2}{r} = \dfrac{(1.74\text{ m/s})^2}{0.200\text{ m}} = 15.1\text{ m/s}^2$.

EVALUATE: $r\omega^2$ and v^2/r are completely equivalent expressions for a_{rad}.

8.

IDENTIFY: $v = r\omega$ and $a_{\tan} = r\alpha$.

SET UP: The linear acceleration of the bucket equals $a_{\tan}$ for a point on the rim of the axle.

EXECUTE: **(a)** $v = R\omega$. $2.00\text{ cm/s} = R\left(\dfrac{7.5\text{ rev}}{\text{min}}\right)\left(\dfrac{1\text{ min}}{60\text{ s}}\right)\left(\dfrac{2\pi\text{ rad}}{1\text{ rev}}\right)$ gives $R = 2.55$ cm.

$D = 2R = 5.09$ cm.

(b) $a_{\tan} = R\alpha$. $\alpha = \dfrac{a_{\tan}}{R} = \dfrac{0.400\text{ m/s}^2}{0.0255\text{ m}} = 15.7\text{ rad/s}^2$.

EVALUATE: In $v = R\omega$ and $a_{\tan} = R\alpha$, ω and α must be in radians.

9.

IDENTIFY: Treat each block as a point mass, so for each block $I = mr^2$, where r is the distance of the block from the axis. The total I for the object is the sum of the I for each of its pieces.

SET UP: In part (a) two blocks are a distance $L/2$ from the axis and the third block is on the axis. In part (b) two blocks are a distance $L/4$ from the axis and one is a distance $3L/4$ from the axis.

EXECUTE: **(a)** $I = 2m(L/2)^2 = \frac{1}{2}mL^2$.

(b) $I = 2m(L/4)^2 + m(3L/4)^2 = \dfrac{1}{16}mL^2(2+9) = \dfrac{11}{16}mL^2$.

EVALUATE: For the same object I is in general different for different axes.

10.

IDENTIFY: I for the object is the sum of the values of I for each part.

SET UP: For the bar, for an axis perpendicular to the bar, use the appropriate expression from Table 9.2. For a point mass, $I = mr^2$, where r is the distance of the mass from the axis.

EXECUTE: **(a)** $I = I_{\text{bar}} + I_{\text{balls}} = \frac{1}{12}M_{\text{bar}}L^2 + 2m_{\text{balls}}\left(\frac{L}{2}\right)^2$.

$$I = \frac{1}{12}(4.00\text{ kg})(2.00\text{ m})^2 + 2(0.500\text{ kg})(1.00\text{ m})^2 = 2.33\text{ kg}\cdot\text{m}^2$$

(b) $I = \frac{1}{3}m_{\text{bar}}L^2 + m_{\text{ball}}L^2 = \frac{1}{3}(4.00\text{ kg})(2.00\text{ m})^2 + (0.500\text{ kg})(2.00\text{ m})^2 = 7.33\text{ kg}\cdot\text{m}^2$

(c) $I = 0$ because all masses are on the axis.

(d) All the mass is a distance $d = 0.500$ m from the axis and

$I = m_{\text{bar}}d^2 + 2m_{\text{ball}}d^2 = M_{\text{Total}}d^2 = (5.00\text{ kg})(0.500\text{ m})^2 = 1.25\text{ kg}\cdot\text{m}^2$.

EVALUATE: I for an object depends on the location and direction of the axis.